Summary in Graph

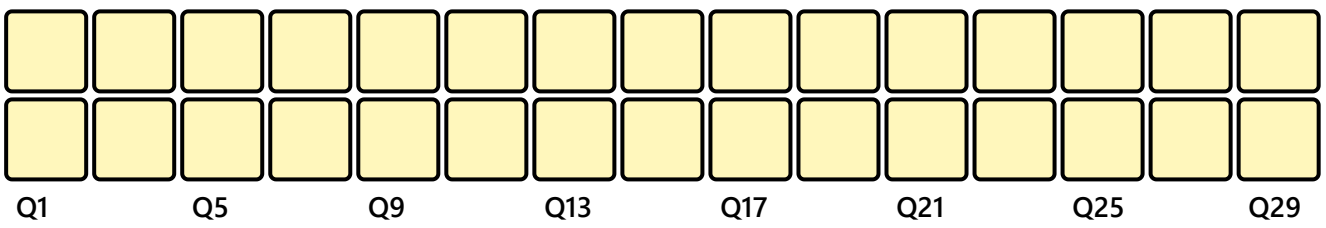
Exam Summary (GATE Overflow CS Test Series 2026 | Probability | Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (3) out of 27 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Probability

A city with 6 districts has 5 robberies in a particular week. Assume the robberies are located randomly, with all possibilities for which robbery occurred where equally likely. What is the probability percentage that some district had more than 1 robbery?

A. more than 90%
B. more than 98%

C. less than 90%
D. less than 98% and more than 97%

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Probability

Consider the following series of numbers

$$1\ 1\ 23\ 43\ 2\ 14\ 7$$

If a denotes the mean, b denotes the mode and c denotes the median of these numbers, $a + 2b + 3c =$ ___

Your Answer: | Correct Answer: 36 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Numerical Type | Award: 1 | Penalty: 0 | Probability

If in a series of 10 values all values are multiplied by 5, the percentage change in the standard deviation will be _____

Your Answer: | Correct Answer: 400 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Which of the following is/are correct? (Mark all the appropriate choices)

A. When n is large, p is small, and $\lambda = np$ is of moderate size, Binomial (n, p) is approximately Poisson (λ)
B. When both n and p are small, Binomial (n, p) is approximately Poisson (λ)
C. When n is small, p is large, and $\lambda = np$ is of moderate size, Binomial (n, p) is approximately Poisson (λ)
D. When both n and p are large, Binomial (n, p) is approximately Poisson (λ)

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Which of the following statements with respect to the random variable X following binomial distribution is/are correct?
Here, p, q and n are the probabilities of success, failure and number of trials respectively.
(Mark all the appropriate choices)

A. Variance of X is given by $np(1-p)$
B. The probability that X takes value i is given by $P(X = i) =^n C_i p^i q^{(n-i)}$
C. The expected value of X is given by np
D. Variance of X is never higher than the expected value of X

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Numerical Type Award: 1 Penalty: 0 Probability

Two dice are rolled. The probability of getting consecutive numbers on both is ___. (Rounded to two decimal points)

Your Answer: Correct Answer: 0.27 : 0.28 Not Attempted Time taken: 00min 00sec Discuss

Q #7 Numerical Type Award: 1 Penalty: 0 Probability

A die is rolled 6 times. If the probability that you get different numbers is $X, 1000X =$ ___ (Rounded to one decimal points)

Your Answer: Correct Answer: 15.4 : 15.5 Not Attempted Time taken: 00min 00sec Discuss

Q #8 Numerical Type Award: 1 Penalty: 0 Probability

A bag has 8 pieces of paper, each with one of the letters of '$\mathrm{GATE2021}$'. If the probability that the letters of the papers picked in order without replacement spell '$\mathrm{GATE2021}$' is $X,\ 10000X =$ ____ (Rounded to two decimal points)

Your Answer: Correct Answer: 0.49 : 0.50 Not Attempted Time taken: 00min 00sec Discuss

Q #9 Multiple Choice Type Award: 1 Penalty: 0.33 Probability

A fair coin is tossed 8 times. What is the probability that we get exactly 6 heads?

A. $\frac{7}{64}$
B. 0.0109
C. $\frac{\binom{8}{6} \times \binom{8}{2}}{2^8}$
D. None of the above

Your Answer: Correct Answer: A Not Attempted Time taken: 00min 00sec Discuss

Q #10 Numerical Type Award: 1 Penalty: 0 Probability

The expected value of the product when two dice are rolled is ____

Your Answer: Correct Answer: 12.25 Not Attempted Time taken: 00min 00sec Discuss

Q #11 Multiple Choice Type Award: 2 Penalty: 0.67 Probability

Assume that there are at least k different people in the room. If the probability that there are two who share a birthday is greater than 10% then which of the following is the least possible value of k. (Assume no one is born in a leap year)

A. 23
B. 12

C. 18
D. 10

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

If three points are randomly chosen along the circumference of a unit circle, dividing the circle into three arcs, the expected length of the arc that contains the point $(1, 0)$ will be ___

A. $\frac{\pi}{3}$
B. $\frac{2\pi}{3}$
C. π
D. $\frac{\pi}{2}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A couple decides to keep having children until they have at least one boy and at least one girl. If the probability for a boy or a girl are the same, and they can produce children indefinitely, what is the expected number of children they will get?

A. 3
B. 2
C. 4
D. 6

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Numerical Type | Award: 2 | Penalty: 0 | Probability

A stick of length $2m$ is broken into two pieces, at a uniformly random chosen break point.The expected length of the larger piece (in meters) is ____

Your Answer: | Correct Answer: 1.5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | Probability

Suppose that on a given weekend the number of accidents at a certain intersection has a poisson distribution with mean 0.6. The probability that there will be at least three accidents in the intersection during the weekend in percentage is ____ (Upto one decimal point)

Your Answer: | Correct Answer: 2.3:2.4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose the probability that some part of an automobile does not work during any one hour period is 0.02, the probability that a given engine will work for 3 hours will be ____

A. 0.96
B. 0.98
C. 0.94
D. 0.88

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 Numerical Type | Award: 2 | Penalty: 0 | Probability

Assume that a random variable X is Normal with mean $\mu = 4$ and variance $\sigma^2 = 16$. The probability that X is between 2 and 6 is ___. (Rounded to 2 decimal points) $\left(\phi(0.5) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{0.5} e^{-x^2/2} dx = 0.6915\right)$

Your Answer: | Correct Answer: 0.38 : 0.39 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 Numerical Type | Award: 2 | Penalty: 0 | Probability

Assume that a cricket ball lasts on average 100 overs. Assuming exponential distribution, the absolute value of the difference in probability that it lasts more than 150 overs and the probability that it lasts less than 80 overs is ___ (correct up to 2 decimal points)

Your Answer: | Correct Answer: 0.32 : 0.33 | Not Attempted | Time taken: 00min 04sec | Discuss

Q #19 Numerical Type | Award: 2 | Penalty: 0 | Probability

In a car factory wings A, B, C manufacture $30, 33$ and 37 percent of the total engines, respectively. Of their output $4, 3$ and 2 percent respectively are defective. If an engine is drawn at random from the produce and is found defective, what is the probability that it was manufactured in the C wing? (Rounded to 2 decimal points)

Your Answer: | Correct Answer: 0.25 : 0.26 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 Numerical Type | Award: 2 | Penalty: 0 | Probability

Professor Alex got the following two series of values as runtimes for a sequence of benchmark runs: $102, 240, 334, 504$ and $105, 225, 380, 510$. Let the geometric mean of the first series be g_1 and that of the second series be g_2. Instead of taking the geometric mean uplift g_u as g_1/g_2, Professor Alex took the geometric mean of the ratios of the two series: $102/105, 240/225, 334/380$ and $504/510$. The percentage deviation of the value he got from g_u is ____ (Rounded to one decimal point)

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

In a series of 2 distinct values, the larger one is increased by 100%. Which of the following statements is/are TRUE? (Mark all the appropriate choices)

A. The arithmetic mean increases by 50%
B. The geometric mean increases by 50%

C. The geometric mean increases by 41.4%
D. The arithmetic mean increases by 41.4%

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Numerical Type | Award: 2 | Penalty: 0 | Probability

Die A has three red and three green faces, whereas die B has four red and two green faces. A fair coin is flipped once. If it falls head, die A is chosen and if it falls tail die B is chosen. The chosen die is thrown repeatedly. If the first 4 throws turn out to be red, what's the probability that die A is being used?

Your Answer: | Correct Answer: 0.24 : 0.25 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Suppose that there are 9 balls in total of colors red and white. The probability that the two randomly chosen balls are of the same color is the same as the probability that they are of different colors. Which of the following is the correct choices for number of red (r) and white (w) balls? (Mark all the appropriate choices)

A. $w = 3, r = 6$
B. $w = 6, r = 3$
C. $w = 4, r = 5$
D. $w = 5, r = 4$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A bag contains one apple which is either green or red, with equal probabilities for both. A green apple is put in the bag (so there are 2 apples now), and then a random apple is taken out. The apple taken out is green. What is the probability that the remaining apple is also green?

A. $\frac{2}{3}$
B. $\frac{1}{3}$
C. $\frac{1}{2}$
D. $\frac{1}{4}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 Numerical Type | Award: 2 | Penalty: 0 | Probability

A spam filter is designed by looking at commonly occurring phrases in spam. Suppose that 70% of email is spam. In 10% of the spam emails, the phrase "online bet" is used, whereas this phrase is only used in 0.5% of non-spam emails. A new email has just arrived, which does mention “online bet”. The probability that it is spam is _____ (Answer upto 2 decimal places)

Your Answer: | Correct Answer: 0.97 : 0.98 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Two coins are in a box. The coins look alike, but one coin is fair (with equal probabilities for Head and Tail), while the other coin is biased, with $1/3$ probability of Heads. One of the coins is randomly chosen from the hat, without knowing which of the two it is. Now this chosen coin is tossed twice, showing Heads both times. Given this information, what is the probability that the chosen coin is the fair coin?

A. $\frac{4}{5}$
B. 0.692
C. $\frac{1}{4}$
D. $\frac{1}{2}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Numerical Type | Award: 2 | Penalty: 0 | Probability

An unbiased coin is tossed repeatedly and its outcome is recorded. What is the expected number of trials to get HT consecutively?

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Which of the following statements is/are TRUE? (Mark all the appropriate choices)

A. Adding a constant to each number in a series increases its arithmetic mean by the same constant
B. Multiplying a constant to each number in a series multiplies its arithmetic mean by the same constant
C. Multiplying a constant to each number in a series multiplies its geometric mean by the same constant
D. Adding a constant to each number in a series increases its standard deviation by the same constant

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Numerical Type | Award: 2 | Penalty: 0 | Probability

A certain family has 8 children, consisting of 3 boys and 5 girls. Assuming that all birth orders are equally likely with no twins, what is the probability that the 3 eldest children are girls? (Up to 2 decimal places)

Your Answer: | Correct Answer: 0.17 : 0.18 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A fair die is thrown as long as necessary for 1 to turn up. If the number of throws (n) turns out to be even, the probability that $n = 4$ is _____

A. $1/36$
B. $5/36$
C. $1/6^4$
D. None of the above

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

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